SOLUTION OF HOMEWORK 05

1.

IDENTIFY: $\tau = Fl$ with $l = r\sin\phi$. Add the two torques to calculate the net torque.
SET UP: Let counterclockwise torques be positive.
EXECUTE: $\tau_1 = -F_1 l_1 = -(8.00\text{ N})(5.00\text{ m}) = -40.0\text{ N}\cdot\text{m}$.
$\tau_2 = +F_2 l_2 = (12.0\text{ N})(2.00\text{ m})\sin 30.0° = +12.0\text{ N}\cdot\text{m}$. $\sum\tau = \tau_1 + \tau_2 = -28.0\text{ N}\cdot\text{m}$. The net torque is $28.0\text{ N}\cdot\text{m}$, clockwise.
EVALUATE: Even though $F_1 < F_2$, the magnitude of τ_1 is greater than the magnitude of τ_2, because F_1 has a larger moment arm.

2.

IDENTIFY: Use $\tau = Fl = rF\sin\phi$ to calculate the magnitude of each torque and use the right-hand rule to determine the direction of each torque. Add the torques to find the net torque.
SET UP: Let counterclockwise torques be positive. For the 11.9 N force (F_1), $r = 0$. For the 14.6 N force (F_2), $r = 0.350\text{ m}$ and $\phi = 40.0°$. For the 8.50 N force (F_3), $r = 0.350\text{ m}$ and $\phi = 90.0°$.
EXECUTE: $\tau_1 = 0$. $\tau_2 = -(14.6\text{ N})(0.350\text{ m})\sin 40.0° = -3.285\text{ N}\cdot\text{m}$.
$\tau_3 = +(8.50\text{ N})(0.350\text{ m})\sin 90.0° = +2.975\text{ N}\cdot\text{m}$. $\sum\tau = -3.285\text{ N}\cdot\text{m} + 2.975\text{ N}\cdot\text{m} = -0.31\text{ N}\cdot\text{m}$. The net torque is $0.31\text{ N}\cdot\text{m}$ and is clockwise.

3.

IDENTIFY: Use $\sum\tau_z = I\alpha_z$ to calculate α. Use a constant angular acceleration kinematic equation to relate α_z, ω_z and t.
SET UP: For a solid uniform sphere and an axis through its center, $I = \frac{2}{5}MR^2$. Let the direction the sphere is spinning be the positive sense of rotation. The moment arm for the friction force is $l = 0.0150\text{ m}$ and the torque due to this force is negative.
EXECUTE: **(a)** $\alpha_z = \dfrac{\tau_z}{I} = \dfrac{-(0.0200\text{ N})(0.0150\text{ m})}{\frac{2}{5}(0.225\text{ kg})(0.0150\text{ m})^2} = -14.8\text{ rad/s}^2$

(b) $\omega_z - \omega_{0z} = -22.5\text{ rad/s}$. $\omega_z = \omega_{0z} + \alpha_z t$ gives $t = \dfrac{\omega_z - \omega_{0z}}{\alpha_z} = \dfrac{-22.5\text{ rad/s}}{-14.8\text{ rad/s}^2} = 1.52\text{ s}$.
EVALUATE: The fact that α_z is negative means its direction is opposite to the direction of spin. The negative α_z causes ω_z to decrease.

4.

IDENTIFY: Apply Eq. (10.8).
SET UP: For an object that is rolling without slipping, $v_{\text{cm}} = R\omega$.
EXECUTE: The fraction of the total kinetic energy that is rotational is

$$\frac{(1/2)I_{\text{cm}}\omega^2}{(1/2)Mv_{\text{cm}}^2 + (1/2)I_{\text{cm}}\omega^2} = \frac{1}{1 + (M/I_{\text{cm}})v_{\text{cm}}^2/\omega^2} = \frac{1}{1 + (MR^2/I_{\text{cm}})}$$

(a) $I_{\text{cm}} = (1/2)MR^2$, so the above ratio is 1/3.
(b) $I_{\text{cm}} = (2/5)MR^2$ so the above ratio is 2/7.
(c) $I_{\text{cm}} = (2/3)MR^2$ so the ratio is 2/5.
(d) $I_{\text{cm}} = (5/8)MR^2$ so the ratio is 5/13.
EVALUATE: The moment of inertia of each object takes the form $I = \beta MR^2$. The ratio of rotational kinetic energy to total kinetic energy can be written as $\dfrac{1}{1 + 1/\beta} = \dfrac{\beta}{1 + \beta}$. The ratio increases as β increases.

5.

IDENTIFY: As the ball rolls up the hill, its kinetic energy (translational and rotational) is transformed into gravitational potential energy. Since there is no slipping, its mechanical energy is conserved.

SET UP: The ball has moment of inertia $I_{cm} = \frac{2}{3}mR^2$. Rolling without slipping means $v_{cm} = R\omega$. Use coordinates where $+y$ is upward and $y = 0$ at the bottom of the hill, so $y_1 = 0$ and $y_2 = h = 5.00$ m. The ball's kinetic energy is $K = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$ and its potential energy is $U = mgh$.

EXECUTE: **(a)** Conservation of energy gives $K_1 + U_1 = K_2 + U_2$. $U_1 = 0$, $K_2 = 0$ (the ball stops).

Therefore $K_1 = U_2$ and $\frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2 = mgh$. $\frac{1}{2}I_{cm}\omega^2 = \frac{1}{2}\left(\frac{2}{3}mR^2\right)\left(\frac{v_{cm}}{R}\right)^2 = \frac{1}{3}mv_{cm}^2$ so

$\frac{5}{6}mv_{cm}^2 = mgh$. Therefore $v_{cm} = \sqrt{\frac{6gh}{5}} = \sqrt{\frac{6(9.80\text{ m/s}^2)(5.00\text{ m})}{5}} = 7.67$ m/s and

$\omega = \frac{v_{cm}}{R} = \frac{7.67\text{ m/s}}{0.113\text{ m}} = 67.9$ rad/s.

(b) $K_{rot} = \frac{1}{2}I\omega^2 = \frac{1}{3}mv_{cm}^2 = \frac{1}{3}(0.426\text{ kg})(7.67\text{ m/s})^2 = 8.35$ J.

EVALUATE: Its translational kinetic energy at the base of the hill is $\frac{1}{2}mv_{cm}^2 = \frac{3}{2}K_{rot} = 12.52$ J. Its total kinetic energy is 20.9 J, which equals its final potential energy:

$mgh = (0.426\text{ kg})(9.80\text{ m/s}^2)(5.00\text{ m}) = 20.9$ J.

6.

IDENTIFY: Apply $P = \tau\omega$ and $W = \tau\Delta\theta$.

SET UP: P must be in watts, $\Delta\theta$ must be in radians, and ω must be in rad/s. 1 rev $= 2\pi$ rad.
1 hp = 746 W. π rad/s = 30 rev/min.

EXECUTE: **(a)** $\tau = \frac{P}{\omega} = \frac{(175\text{ hp})(746\text{ W/hp})}{(2400\text{ rev/min})\left(\frac{\pi}{30}\frac{\text{rad/s}}{\text{rev/min}}\right)} = 519\text{ N}\cdot\text{m}.$

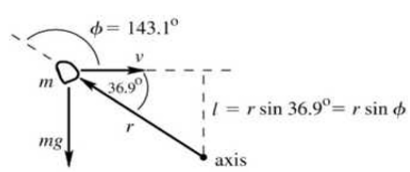
(b) $W = \tau\Delta\theta = (519\text{ N}\cdot\text{m})(2\pi\text{ rad}) = 3260$ J

EVALUATE: $\omega = 40$ rev/s, so the time for one revolution is 0.025 s. $P = 1.306\times10^5$ W, so in one revolution, $W = Pt = 3260$ J, which agrees with our result.

7.

(a) IDENTIFY: Use $L = mvr\sin\phi$ (Eq. (10.25)):

SET UP: Consider Figure 10.37.



EXECUTE: $L = mvr\sin\phi =$
$(2.00\text{ kg})(12.0\text{ m/s})(8.00\text{ m})\sin 143.1^\circ$
$L = 115\text{ kg}\cdot\text{m}^2/\text{s}$

To find the direction of $\vec{L}$ apply the right-hand rule by turning $\vec{r}$ into the direction of $\vec{v}$ by pushing on it with the fingers of your right hand. Your thumb points into the page, in the direction of $\vec{L}$.

(b) IDENTIFY and **SET UP:** By Eq. (10.26) the rate of change of the angular momentum of the rock equals the torque of the net force acting on it.

EXECUTE: $\tau = mg(8.00\text{ m})\cos 36.9^\circ = 125\text{ kg}\cdot\text{m}^2/\text{s}^2$

To find the direction of $\vec{\tau}$ and hence of $d\vec{L}/dt$, apply the right-hand rule by turning $\vec{r}$ into the direction of the gravity force by pushing on it with the fingers of your right hand. Your thumb points out of the page, in the direction of $d\vec{L}/dt$.

EVALUATE: $\vec{L}$ and $d\vec{L}/dt$ are in opposite directions, so L is decreasing. The gravity force is accelerating the rock downward, toward the axis. Its horizontal velocity is constant but the distance l is decreasing and hence L is decreasing.

8.

IDENTIFY and **SET UP:** There is no net external torque about the rotation axis so the angular momentum $L = I\omega$ is conserved.

EXECUTE: **(a)** $L_1 = L_2$ gives $I_1\omega_1 = I_2\omega_2$, so $\omega_2 = (I_1/I_2)\omega_1$

$I_1 = I_{tt} = \frac{1}{2}MR^2 = \frac{1}{2}(120\text{ kg})(2.00\text{ m})^2 = 240\text{ kg}\cdot\text{m}^2$

$I_2 = I_{tt} + I_p = 240\text{ kg}\cdot\text{m}^2 + mR^2 = 240\text{ kg}\cdot\text{m}^2 + (70\text{ kg})(2.00\text{ m})^2 = 520\text{ kg}\cdot\text{m}^2$

$\omega_2 = (I_1/I_2)\omega_1 = (240\text{ kg}\cdot\text{m}^2/520\text{ kg}\cdot\text{m}^2)(3.00\text{ rad/s}) = 1.38\text{ rad/s}$

(b) $K_1 = \frac{1}{2}I_1\omega_1^2 = \frac{1}{2}(240\text{ kg}\cdot\text{m}^2)(3.00\text{ rad/s})^2 = 1080\text{ J}$

$K_2 = \frac{1}{2}I_2\omega_2^2 = \frac{1}{2}(520\text{ kg}\cdot\text{m}^2)(1.38\text{ rad/s})^2 = 495\text{ J}$

EVALUATE: The kinetic energy decreases because of the negative work done on the turntable and the parachutist by the friction force between these two objects.
The angular speed decreases because I increases when the parachutist is added to the system.

9.

IDENTIFY: Use Eq. (11.3) to calculate x_{cm}. The center of gravity of the bar is at its center and it can be treated as a point mass at that point.

SET UP: Use coordinates with the origin at the left end of the bar and the $+x$ axis along the bar.

$m_1 = 0.120\text{ kg},\ m_2 = 0.055\text{ kg},\ m_3 = 0.110\text{ kg}.$

EXECUTE: $x_{cm} = \dfrac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3} = \dfrac{(0.120\text{ kg})(0.250\text{ m}) + 0 + (0.110\text{ kg})(0.500\text{ m})}{0.120\text{ kg} + 0.055\text{ kg} + 0.110\text{ kg}} = 0.298\text{ m}.$ The fulcrum should be placed 29.8 cm to the right of the left-hand end.

EVALUATE: The mass at the right-hand end is greater than the mass at the left-hand end. So the center of gravity is to the right of the center of the bar.

10.

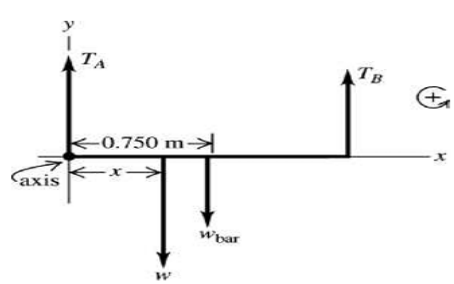
IDENTIFY: Apply the conditions for equilibrium to the bar. Set each tension equal to its maximum value.

SET UP: Let cable A be at the left-hand end. Take the axis to be at the left-hand end of the bar and x be the distance of the weight w from this end. The free-body diagram for the bar is given in Figure 11.9.

EXECUTE: **(a)** $\Sigma F_y = 0$ gives $T_A + T_B - w - w_{bar} = 0$ and

$w = T_A + T_B - w_{bar} = 500.0\text{ N} + 400.0\text{ N} - 350.0\text{ N} = 550\text{ N}.$

(b) $\Sigma\tau_z = 0$ gives $T_B(1.50\text{ m}) - wx - w_{bar}(0.750\text{ m}) = 0.$

$x = \dfrac{T_B(1.50\text{ m}) - w_{bar}(0.750\text{ m})}{w} = \dfrac{(400.0\text{ N})(1.50\text{ m}) - (350\text{ N})(0.750\text{ m})}{550\text{ N}} = 0.614\text{ m}.$ The weight should be placed 0.614 m from the left-hand end of the bar (cable A).

EVALUATE: If the weight is moved to the left, T_A exceeds 500.0 N and if it is moved to the right T_B exceeds 400.0 N.